Date : 09/03/2026

Question – Phone Book

Solution :

```
#include <stdio.h>
#include <string.h>

#define MAX 20
struct Friend {
   char name[30];
   char phone[15];
   char location[50];
   char coming;  /* Y or N */
};
struct Friend pb[MAX];
int n = 0;
void add() {
   printf("Name     : "); scanf(" %[^\n]", pb[n].name);
   printf("Phone    : "); scanf(" %[^\n]", pb[n].phone);
   printf("Location : "); scanf(" %[^\n]", pb[n].location);
   pb[n].coming = '?';
   n++;
   printf("Added!\n");
}
void send_invite() {
   for (int i = 0; i < n; i++)
      printf("Invitation sent to %s (%s) - Hey %s, you are invited to my birthday party!\n",
            pb[i].name, pb[i].phone, pb[i].name);
}

void mark_attendance() {
   for (int i = 0; i < n; i++) {
```

```
        char c;
        printf("Is %s coming? (y/n): ", pb[i].name);
        scanf(" %c", &c);
        pb[i].coming = (c == 'y' || c == 'Y') ? 'Y' : 'N';
    }
}
void view() {
    printf("\n%-20s %-15s %-20s %s\n", "NAME", "PHONE", "LOCATION", "STATUS");
    puts("------------------------------------------------------------");
    for (int i = 0; i < n; i++)
        printf("%-20s %-15s %-20s %s\n",
               pb[i].name, pb[i].phone, pb[i].location,
               pb[i].coming == 'Y' ? "Coming" :
               pb[i].coming == 'N' ? "Not Coming" : "Pending");
    puts("------------------------------------------------------------");
}
int main() {
    int c;
    printf("=== Birthday Phonebook ===\n");
    do {
        printf("\n1.Add Friend  2.Send Invites  3.Mark Attendance  4.View List  5.Exit\nChoice: ");
        scanf("%d", &c);
        if (c==1) add();
        else if (c==2) send_invite();
        else if (c==3) mark_attendance();
        else if (c==4) view();
        else if (c==5) printf("Bye!\n");
        else printf("Invalid!\n");
    } while (c != 5);
    return 0;
}
```

1. **Explanation :**

   This program is a Birthday Phonebook Management System.
   It is used to store my friends' details and manage birthday party invitations.

2. **I used a structure called Friend.**
   This structure stores four details of each friend:

   1)Name

   2)Phone number+

   3)Location

   4)Coming status (Y / N / Pending)

3. **I created an array of the structure** called pb[MAX].
   This allows the program to store **up to 20 friends** in the phonebook.

4. **The variable n is used to track the number of friends** currently stored in the list.

5. **Add Function (add())**
   This function is used to add a new friend.
   It asks the user to enter the friend's name, phone number, and location.
   Initially the coming status is set to **'?'**, which means attendance is not yet confirmed.

6. **Send Invitation Function (send_invite())**
   This function sends a birthday invitation message to all stored friends.
   It uses a **for loop** to go through each friend and prints an invitation message.

7. **Mark Attendance Function (mark_attendance())**
   This function asks whether each friend is coming to the party or not.
   If the user enters **y**, the status becomes **Coming (Y)**.
   If the user enters **n**, the status becomes **Not Coming (N)**.

8. **View Function (view())**
   This function displays the complete phonebook list.
   It shows **Name, Phone, Location, and Status** in a formatted table.

9. **Main Function**
   The main function provides a **menu-driven interface** with options:

   1)Add Friend

   2)Send Invites

   3)Mark Attendance

   4)View List

   5)Exit

10. **A do-while loop is used** so the program keeps running until the user selects **Exit**.

11. **Concepts used in this program**

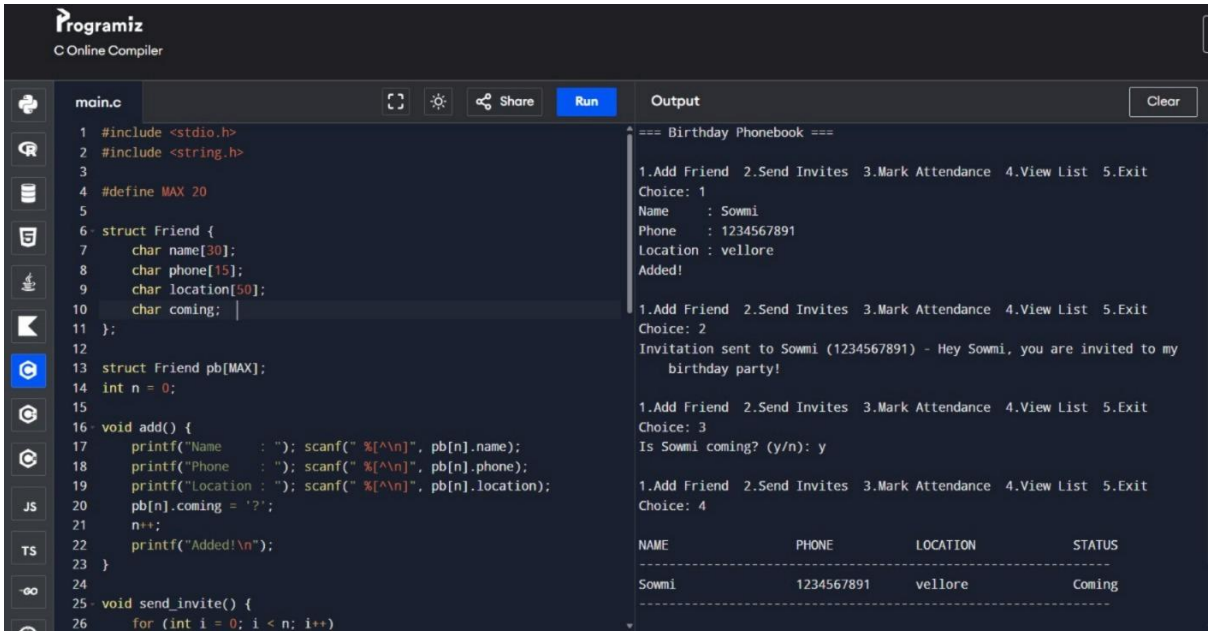
    1)Structures

    2)Arrays of structures

3)Functions

4)Loops

)Conditional statements

Finally this program helps organize friends details and track who will attend the birthday party or not.

Output: